

The Art of Computer Programming 3

Lecture Notes

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Lecture 1. Computable and uncomputable functions

1.1 Alphabets and words

At the low level, information inside computers is represented by 0s and 1s. Every program takes as input a finite sequence of bits and returns another such sequence, and all computation as we know it, in the end, is manipulation of bit-strings. Our formalisation begins here.

Definition 1 (Alphabets and words). An **alphabet** Σ is a non-empty finite set of symbols. A **word** (or **string**) over Σ is a finite sequence of symbols $w = a_1 a_2 \dots a_n$, where each $a_i \in \Sigma$. The unique sequence of length zero is the **empty word**, denoted ε . The set of all finite words over Σ is denoted Σ^* .

Given two words $u = a_1 \dots a_n$ and $v = b_1 \dots b_m$, their **concatenation** is $uv = a_1 \dots a_n b_1 \dots b_m$. Concatenation is associative: $(uv)w = u(vw)$, and ε is the two-sided identity: $u\varepsilon = \varepsilon u = u$.¹

Binary encoding. The ‘canonical’ alphabet for computers is the *binary alphabet* $\{0, 1\}$. Words in any other alphabet Σ can be *encoded* by binary words as follows. We fix a length k such that $2^k \geq |\Sigma|$, and assign a distinct k -bit binary word to each symbol a_i . For example, if $\Sigma = \{a, b, c\}$, we can choose $k = 2$ and map

$$a \mapsto 00, \quad b \mapsto 01, \quad c \mapsto 10.$$

A word over Σ is then encoded by replacing each symbol with its code: for instance, cab becomes 100001. Since every symbol uses exactly 2 bits (or, in general, k), decoding is unambiguous: we simply read the string two bits at a time.²

Other finite objects that programs work with are encoded similarly. For example, a directed graph G on vertices $\{v_1, \dots, v_n\}$ can be represented by its $n \times n$ adjacency matrix A , where $A_{i,j} = 1$ if there is an edge $v_i \rightarrow v_j$ and 0 otherwise. Flattening this matrix row by row yields a binary word of length n^2 . For instance, the graph on 3 vertices with edges $v_1 \rightarrow v_2$ and $v_2 \rightarrow v_3$ and its corresponding adjacency matrix are:



$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

which is encoded by the word 010001000.

1.2 Computable functions

Fix your favourite general-purpose programming language (we will use Python in these notes). Consider programs that take one input word and produce one output word over some alphabet Σ . As explained above, multiple arguments, numbers, graphs, or structured data can always be encoded into a single word in Σ^* (and decoded from the output).

¹If you like algebra, you may note that Σ^* with concatenation and ε is thus the free monoid generated by Σ .

²In ASCII, a standard encoding in real computers, each character is an 8-bit byte.

Thus, each program π naturally computes a (partial!) function $f_\pi: \Sigma^* \rightharpoonup \Sigma^*$.³ This is how it is defined.

Definition 2 (Function computed by a program π). *Given $u \in \Sigma^*$, run π on u .*

- *If π halts (i.e., terminates, concludes the execution), then $f_\pi(u) = v$, where $v \in \Sigma^*$ is the word it outputs.*
- *If π does not halt, then $f_\pi(u)$ is undefined.*

In real life, programs may also crash with an error. In theory, we treat crashes as equivalent to non-halting.

Definition 3 (Computable function). *A partial function $f: \Sigma^* \rightharpoonup \Sigma^*$ is called **computable** if there exists a program π such that $f = f_\pi$.*

Note that when $f: \Sigma^* \rightarrow \Sigma^*$ is **total**, i.e., defined on all words in Σ^* , the corresponding program π must halt on every input.

Let us consider some examples of computable functions.

Example 4. The following functions are all computable:

- *reverse*: given $w \in \Sigma^*$, outputs its reverse (e.g., $\text{reverse}(abc) = cba$);
- *convert*: interprets $w \in \{0, 1, \dots, 9\}^*$ as an integer n , outputs a word in $\{0, 1, 2, 3\}^*$ —the same number, but in 4-ary notation;
- *double*: interprets w as an integer n , outputs $2n$;
- *multiply*: given a pair of integers $\langle n, k \rangle$, outputs their product $n \cdot k$;
- *shortestPath*: given a directed graph and two vertices $\langle G, s, t \rangle$, outputs the length of the shortest path from s to t (undefined if no path exists);
- *deanon*: given w , if w encodes the real name of Banksy, $\text{deanon}(w)$ encodes the real name of Satoshi Nakamoto; for all other w , *deanon* is undefined.

1.3 Uncomputable functions

Are there any functions that are not computable? Yes, and it is easy to prove: there are only countably many programs, but uncountably many functions on $\{0, 1\}^*$. Let us see this argument in detail.

³Note the arrow! It is $\rightharpoonup$, not $\rightarrow$. This is how we denote that the function is *partial*.

Recall that a set S is called *countable* if its elements can be indexed by natural numbers. The set $\{0, 1\}^*$ is, of course, countable—here is a possible indexing of words in $\{0, 1\}^*$:

$$\begin{aligned}
 w_0 &= \varepsilon \\
 w_1 &= 0 \\
 w_2 &= 1 \\
 w_3 &= 00 \\
 w_4 &= 01 \\
 w_5 &= 10 \\
 w_6 &= 11 \\
 w_7 &= 000 \\
 &\dots
 \end{aligned} \tag{1}$$

That is, we list the words by increasing length, and order words of the same length lexicographically. Similarly, Σ^* is countable for any alphabet Σ .

Programs are words themselves. Indeed, programs are just sequences of symbols that we write, and thus can be seen as words. In fact, the converse is also true: we can try to execute any word as if it was a program and see what happens—a crash would just mean non-halting. So every program is a word and every word encodes some (maybe nonsensical) program. The set of all programs, and hence of all computable functions, is therefore countable.

In contrast, the set of all partial functions $\{0, 1\}^* \rightarrow \{0, 1\}^*$ is uncountable. In fact, this is already true for the set of all functions $\{0, 1\}^* \rightarrow \{0, 1\}$.

Proposition 5 (Cantor, 1891). *The set of all functions $\{0, 1\}^* \rightarrow \{0, 1\}$ is uncountable.*

Proof. For clarity of notation, we use the correspondence (1) and consider the set of functions $\mathbb{N} \rightarrow \{0, 1\}$. Assume for contradiction that this set is countable. Then we can index all functions in this set: $f_0, f_1, f_2, \dots$. Each function f_i is defined by its values on each $k \in \mathbb{N}$. Let us write them out:

$f_0(0)$	$f_0(1)$	$f_0(2)$	$f_0(3)$	$\dots$
$f_1(0)$	$f_1(1)$	$f_1(2)$	$f_1(3)$	$\dots$
$f_2(0)$	$f_2(1)$	$f_2(2)$	$f_2(3)$	$\dots$
$f_3(0)$	$f_3(1)$	$f_3(2)$	$f_3(3)$	$\dots$
$\dots$				

Consider the diagonal of this infinite table, in bold, and the function:

$$d(n) = \begin{cases} 1, & \text{if } f_n(n) = 0, \\ 0, & \text{if } f_n(n) = 1. \end{cases}$$

Then, d is a function $\mathbb{N} \rightarrow \{0, 1\}$, so it must appear in the above list $f_0, f_1, f_2, \dots$ of all such functions. However, by construction, d disagrees with f_0 on 0, with f_1 on 1, and, in general, with f_n on n . A contradiction. $\square$

Corollary 6. *There exist uncomputable functions from $\{0,1\}^*$ to $\{0,1\}$.*

In fact, there are uncountably many.

And so what? No concrete example was given so far. Could it be that all *useful* functions that we may ever want to compute are countable? Unfortunately, not. Consider the following very useful *halting function*:

$$h(\langle \pi, w \rangle) = \begin{cases} 1, & \text{if } \pi \text{ halts on input } w, \\ 0, & \text{if } \pi \text{ loops forever on input } w. \end{cases}$$

Its argument is a *pair*: a program π and a word w , put in the $\langle \rangle$ -brackets and separated by a comma—this is just a way to ‘pack’ two different things into one argument of a function. We assume that $\langle, \rangle$ and the comma are all symbols of Σ that are not used in programs and input words.⁴ Note that h is a total function.

Theorem 7 (Turing, 1936). *The halting predicate is (alas!) uncomputable.*

Proof. Recall that we have an indexing of words in Σ^* : $w_0, w_1, w_2, \dots$, each of which can be seen as a program: $\pi_0, \pi_1, \pi_2, \dots$. Each π_i can (potentially) be run on each w_j (if $i = j$, we run π_i on its own code!)

We can imagine a table of results for the halting predicate:

	w_0	w_1	w_2	w_3	$\dots$
π_0	$h(\pi_0, w_0)$	$h(\pi_0, w_1)$	$h(\pi_0, w_2)$	$h(\pi_0, w_3)$	$\dots$
π_1	$h(\pi_1, w_0)$	$h(\pi_1, w_1)$	$h(\pi_1, w_2)$	$h(\pi_1, w_3)$	$\dots$
π_2	$h(\pi_2, w_0)$	$h(\pi_2, w_1)$	$h(\pi_2, w_2)$	$h(\pi_2, w_3)$	$\dots$
π_3	$h(\pi_3, w_0)$	$h(\pi_3, w_1)$	$h(\pi_3, w_2)$	$h(\pi_3, w_3)$	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

We are going to mess with the diagonal again! Suppose for contradiction that h is computed by some program η . We define the following *diagonal* program δ :

```
def  $\delta(w)$ :
    if  $\eta(w, w) == 1$ :
        while True: pass    # loop forever
    else:
        return 0             # halt immediately
```

Since δ is a valid program, we could ask what happens when δ is run on its own code. If δ halts when run on the word encoding δ , then $\eta(\delta, \delta)$ must return 1, and so δ , according to its own code, should actually loop forever. If it does not halt, $\eta(\delta, \delta)$ returns 0, and then δ should halt. A contradiction. $\square$

⁴This creates a problem: we need to define h also on ‘malformed’ inputs—such as $\langle \langle, \rangle, \rangle$ —which is formally a word from Σ^* . But we can agree that h will be equal to 0 on such words and forget about them. What we really care about is the behaviour of h on properly formed words: a $\langle$, then a program, then a comma, then an input word, then a $\rangle$.

Lecture 2. Decidable and enumerable languages

2.1 From functions to languages

Let $f: \Sigma^* \rightarrow \{0, 1\}$ be a total function that outputs a single bit. It can be thought of as classifying input words into two categories: those on which f returns 1 (the ‘yes’ words) and those on which f returns 0 (the ‘no’ words).

This is the same as specifying a subset $L \subseteq \Sigma^*$ —namely, the set $L = \{w \in \Sigma^* \mid f(w) = 1\}$.

Conversely, every set $L \subseteq \Sigma^*$ defines a one-bit function, called its **characteristic function**:

$$\chi_L(w) = \begin{cases} 1, & \text{if } w \in L, \\ 0, & \text{if } w \notin L. \end{cases}$$

Every set $L \subseteq \Sigma^*$ is associated with the following problem of binary classification, called the **decision problem**⁵ for L :

Input: $w \in \Sigma^*$

Question: Does w belong to L ?

Definition 8. A set of words $L \subseteq \Sigma^*$ is called a **language**. It is **decidable** when its characteristic function is computable, and **undecidable** otherwise.

A program that computes the characteristic function χ_L is said to **decide** the language L : it outputs 1 on words from L (accepts them) and 0 on words not from L (rejects them). Note that such a program must halt on every input—and give an answer.

We thus have a correspondence between one-bit functions and languages:

	Function $f: \Sigma^* \rightarrow \{0, 1\}$	Language $L \subseteq \Sigma^*$
Task (‘problem’)	given w , compute $f(w)$	given w , decide whether $w \in L$
If some π does that	computable	decidable
This π	computes the function	decides the language

Example 9. The empty language $L = \emptyset$ is decidable. It is decided by the following program:

```
def reject_all(w):
    return 0
```

Example 10. Any finite language $L = \{w_0, w_1, \dots, w_n\}$ is decidable:

⁵The term is historical: basically, we are asked to ‘decide’ if the input word belongs to the set.

```
def decide_L(w):
    words_L = [w0, w1, ... , wn]
    if w in words_L:
        return 1
    else:
        return 0
```

Example 11. The language

$$\text{PRIMES} = \{w \in \{0,1\}^* \mid w \text{ is the binary encoding of a prime number}\}$$

is decidable:

```
def is_prime(w):
    n = int(w, 2)          # interpret binary string as integer
    if n < 2:
        return 0
    for d in range(2, n):
        if n % d == 0:
            return 0
    return 1
```

Example 12. The language

$$\text{HALT} = \{\langle \pi, w \rangle \mid \pi \text{ halts on input } w\}$$

is undecidable, by Theorem 7.

It is a tradition in the theory of computing to name languages in ALL CAPS. It might be slightly ANNOYING, but that's the tradition, and we will RESPECT IT.

2.2 Semantic properties of programs

We have seen that HALT is undecidable, which is sad for those who would like to *verify* that their programs behave in the expected manner—at least, halt on a given word. Are there properties of programs that one can effectively check?

Of course there are. For example, this one:

Input: program code

Question: Does this program use a variable named x ?

It should be clear that the property above is qualitatively different from HALT. The former asks about the program's *syntax*, the latter—about the *function* that the program computes, i.e. the program's *semantics*. To highlight this distinction, we introduce some terminology.

Recall that every $w \in \Sigma^*$ can be treated as a program: feed it to Python's interpreter, specify an input, and it will do something—maybe crash immediately, maybe output Banksy's real name, or maybe output the input backwards...—it will compute some partial function $\Sigma^* \rightarrow \Sigma^*$.

Any language $P \subseteq \Sigma^*$ defines a **property of programs**: a program π has property P if $\pi \in P$, and does not have it if $\pi \notin P$.

Definition 13. A property of programs $P \subseteq \Sigma^*$ is called **semantic** if for any two programs $\pi_1, \pi_2 \in \Sigma^*$, if $f_{\pi_1} = f_{\pi_2}$, then either both $\pi_1, \pi_2 \in P$ or both $\pi_1, \pi_2 \notin P$.

In other words, a semantic property is a property of *functions*: for every function, either all programs that compute it have the property, or none does.

A property P is **non-trivial** if $P \neq \emptyset$ and $P \neq \Sigma^*$. That is, there are some programs that have it, and some that do not.

Example 14. Consider the property

$$\text{HALT-ON-EMPTY} = \{\pi \mid \pi \text{ halts on input } \varepsilon\}.$$

It is semantic: $\pi \in \text{HALT-ON-EMPTY}$ iff $f_\pi(\varepsilon)$ is defined. It is non-trivial: a program that always outputs 1 regardless of the input has this property, and one that loops forever on all inputs does not have it.

The following theorem is the saddest theorem in our course.

Theorem 15 (Rice, 1953). *Every non-trivial semantic property of programs is undecidable.*

Proof. The proof was not covered in the lecture, so it is here just for the curious.

Let $P \subseteq \Sigma^*$ be a non-trivial semantic property, and consider the following program:

```
def never_stop(w):
    while True: pass    # loops forever regardless of the input
```

The function that it computes, $f_{\text{never_stop}}$, is the (unique) function that is undefined on all inputs.

Suppose that $\text{never_stop} \notin P$. (If $\text{never_stop} \in P$, consider the complement $\bar{P} = \Sigma^* \setminus P$ instead, which is also a non-trivial semantic property. Properties P and $\bar{P}$ are either both decidable or both undecidable, so this is without loss of generality.)

Since P is non-empty, there is some program $\sigma \in P$. Since P is semantic and $\text{never_stop} \notin P$, we have that $f_\sigma \neq f_{\text{never_stop}}$, meaning that σ halts on at least one input.

We reduce the Halting Problem to P . Given a pair $\langle \pi, w \rangle$, construct the following program $\rho_{\pi, w}$:

```
def rho_pi_w(x):
    pi(w)                # run pi on w (ignore x)
    return sigma(x)      # if pi halted, behave like sigma
```

If π halts on w , then $\rho_{\pi,w}$ behaves exactly like σ , so $f_{\rho_{\pi,w}} = f_\sigma$ and $\rho_{\pi,w} \in P$. If π does not halt on w , then $\rho_{\pi,w}$ never reaches the second line, so $f_{\rho_{\pi,w}} = f_{\text{never_stop}}$ and $\rho_{\pi,w} \notin P$.

Therefore, $\langle \pi, w \rangle \in \text{HALT}$ if and only if $\rho_{\pi,w} \in P$. If P were decidable by some program `decide_P`, we would be able to decide HALT as follows:

```
def decide_halt( $\pi$ , w):
     $\rho_{\pi,w}$  = construct( $\pi$ , w)    # construct the code of  $\rho_{\pi,w}$  as above
    return decide_P( $\rho_{\pi,w}$ )      # check whether  $\rho_{\pi,w} \in P$ 
```

This contradicts Theorem 7. □

Example 16. The following languages are all undecidable by Rice’s theorem:

- $\{\pi \mid \pi \text{ halts on all inputs}\}$;
- $\{\pi \mid \pi \text{ halts on some input}\}$;
- $\{\pi \mid \pi \text{ accepts the empty word}\}$;
- $\{\pi \mid f_\pi = f_{\pi_0}\}$ for any fixed π_0 — the set of programs equivalent to π_0 .

Remark 17. There is one hidden detail in the proof of Rice’s theorem—the *construction* of the program $\rho_{\pi,w}$ from π and w . For a proper reduction, we need this construction to be computable: given the code of π and the word w , we must be able to algorithmically produce the code of $\rho_{\pi,w}$. In Python (and any real programming language), this is obvious: $\rho_{\pi,w}$ is just a piece of source code assembled by string concatenation. Formally, a programming language (‘encoding’) is *admissible* if it has an interpreter and ensures the ability to algorithmically hard-code arguments into a given program’s source code. Rice’s theorem holds for any admissible encoding, but can fail in artificial, pathological programming languages where programs cannot be algorithmically constructed (imagine, for example, a language where *somehow* every program has an additional flag—whether it halts on the empty input or not).

2.3 Enumerable languages

Definition 18. A language $L \subseteq \Sigma^*$ is called **recursively enumerable** (or simply **enumerable**) if there exists a program that runs indefinitely and prints a (possibly infinite) list of words, such that:

- (1) every word printed by the program belongs to L ;
- (2) every word belonging to L is eventually printed by the program (possibly with repetitions).

Example 19. The set $\text{EVEN} = \{n \in \mathbb{N} \mid n = 2k, k \in \mathbb{N}\}$ is enumerable:

```
def enum_even():
    x = 0
    while True:
        print(x)
        x = x + 2
```

Example 20. The language $\{0, 1\}^*$ is enumerable:

```
def enum_bin_words():
    words = [""]      # create an array with just ε
    for w in words:
        print(w)
        words.append(w + "0")
        words.append(w + "1")
```

If we replace `print(w)` with `yield w`, `enum_bin_words()` becomes an infinite generator, allowing us to write `for w in enum_bin_words():` to iterate over all words in $\{0, 1\}^*$ just like a Python `range`.

By analogy, Σ^* is enumerable and can be iterated through, for any alphabet Σ .

Example 21. The language $\text{HALT} = \{\langle \pi, w \rangle \mid \pi \text{ halts on input } w\}$ is enumerable.

Since $\{0, 1\}^*$ is enumerable, we can enumerate (binary encodings of) all programs and all input words. But to enumerate the pairs $\langle \pi, w \rangle \in \text{HALT}$, we would need to know if π halts on w . How can we tell, if HALT is undecidable?

Imagine we had infinitely many processors, one for each pair $\langle \pi_i, w_j \rangle$. Then, for every $\langle \pi_i, w_j \rangle$ we can run a process p_{ij} , all of them in parallel, and wait. When a process terminates, we print the respective pair to the output. And those processes that never terminate will never have their pairs printed—that's exactly what we want.

The hard part is how to simulate infinitely many processes on a normal computer. We will rely on the following consideration. At the lowest level, a program does not execute all at once: its execution proceeds as a discrete sequence of elementary *machine instructions*—simple primitive steps: fetching a byte from memory, performing an arithmetic operation, or jumping to another location in code. Each instruction corresponds to a single unit of execution time.

So we can simulate the ‘parallelisation’ as follows. First, we arrange the infinite two-dimensional grid of processes p_{ij} into a single linear sequence: $p_0, p_1, p_2, \dots$ (for instance, ordering by the sum $i + j$). Then, we share the processor by gradually allocating more instructions to more processes:

- On step 0, run p_0 for 1 instruction.
- On step 1, run p_0 for a 2nd instruction, and p_1 for 1 instruction.
- On step 2, run p_0 for a 3rd instruction, p_1 for a 2nd instruction, and p_2 for 1 instruction.
- In general, on step s , run p_i for its $(s - i + 1)$ -th instruction for each $i \in \{0, 1, \dots, s\}$.

Our program `enum_halt` will maintain a list of running processes and for each store their current progress. It will iterate through this list, pushing each process forward by one step, and appending newly launched processes. When a process terminates, the respective pair $\langle \pi_i, w_j \rangle$ is printed out.

This technique has a vintage name, *dovetailing*. The metaphor comes from carpentry: a *dovetail joint* joins two boards by interlocking alternating fingers of wood. Here, we similarly interleave small slices of execution from different processes.

Theorem 22 (Post, 1944). *A language $L \subseteq \Sigma^*$ is decidable if and only if both L and its complement $\bar{L} = \Sigma^* \setminus L$ are enumerable.*

Proof. ($\Rightarrow$) Suppose L is decided by a program `decide_L`. Let `enum_words()` be a generator for Σ^* , analogous to the `enum_bin_words` from Example 20. We can enumerate L simply by filtering:

```
def enum_L():
    for w in enum_words():          # iterate over all words in  $\Sigma^*$ 
        if decide_L(w) == 1:
            print(w)
```

For the complement $\bar{L}$, we do the exact same thing, replacing `== 1` with `== 0`.

($\Leftarrow$) Let `enum_L` and `enum_not_L` enumerate L and $\bar{L}$, respectively. On input w , run `enum_L` and `enum_not_L` in parallel until one of them outputs w (on two processors, or on one, via dovetailing). If `enum_L` outputs w , return 1; if `enum_not_L` outputs w , return 0. Since w must belong to either L or $\bar{L}$ (and never both), exactly one will output w at some point, so this decides L . $\square$

Corollary 23. *The complement of the Halting Problem,*

$$\overline{\text{HALT}} = \{\langle \pi, w \rangle \mid \pi \text{ does not halt on } w\}$$

*is **not** enumerable.*

Lecture 3. Turing machines

3.1 The 1930s: What is computation?

Until now, we have used Python to write programs—our statement of Turing’s theorem is that no Python program computes the halting function (in the terminology of Lecture 1) or decides the language HALT (in the terminology of Lecture 2). However, Turing proved his theorem in 1936, before electronic programmable computers existed and when pythons were more dangerous to work with. This deserves an explanation.

The story begins in the 1920s with David Hilbert’s grand programme (not program!) to establish the foundations of mathematics. A central question of this programme was the *Entscheidungsproblem* (the ‘decision problem’), formulated by Hilbert and Wilhelm Ackermann in 1928:

Does there exist an effective procedure that takes as input any mathematical statement (encoded in a special notation called ‘first-order logic’) and decides whether the statement is true or false?

No electronic computers existed at the time, so it was supposed that such a ‘procedure’ would be executed by a person with pen and paper. For a procedure to be ‘effective’, it must be applicable mechanically, by following clear instructions step by step, without creative thinking or other usual attributes of a mathematician’s work.

Example 24. Consider solving a quadratic equation $ax^2 + bx + c = 0$, where $a \neq 0$, over the real numbers. Compute the discriminant $\Delta = b^2 - 4ac$. If $\Delta < 0$, there are no real solutions; otherwise, evaluate $x = \frac{-b \pm \sqrt{\Delta}}{2a}$. This is an effective procedure: the steps are explicit and require no mathematical insight or intuition.

Hilbert believed mathematics was decidable: that any well-posed mathematical question could be decided mechanically. To prove this belief rigorously—or to disprove it, as eventually happened—one needed a formal definition of an effective procedure, i.e. of an *algorithm*.

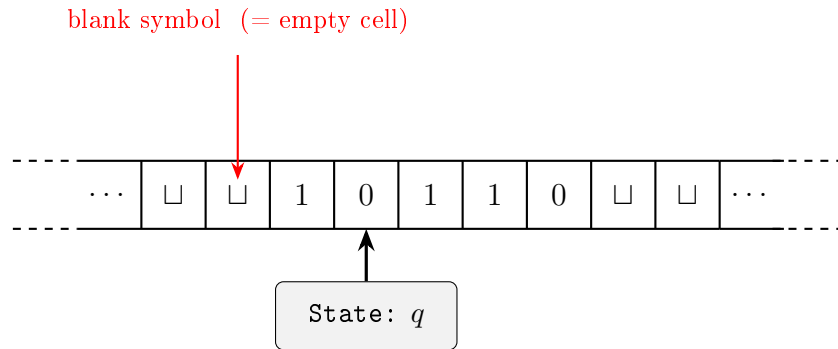
Several definitions of an algorithm were proposed in the 1930s. Although developed by different people, they all turned out to be equivalent in terms of which functions are computable. This is strong evidence that they capture the ‘nature of computation’. We will consider some of these definitions in our course.

Probably the most famous definition, and the first that we will study (although not the first historically), was given by Alan Turing in 1936.

3.2 The Turing machine

Informally speaking, a Turing machine is a device consisting of three components:

1. An **infinite tape** divided into discrete cells, extending indefinitely in both directions. Each cell can hold exactly one symbol from a finite **tape alphabet** Γ .
2. A **read/write head** that points to a single cell on the tape at any moment.
3. A **finite control** that stores the internal state q from a finite set of states Q .



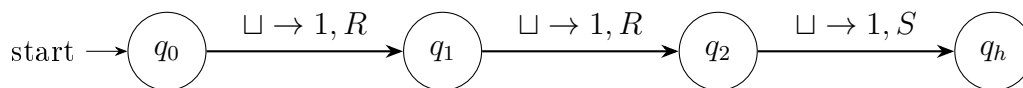
The machine works in steps. At each step, it reads the symbol written in the current cell, and then, depending on that symbol and on the state it is in:

1. overwrites the cell with a new symbol;
2. moves one cell to the right, or to the left, or stays put;
3. switches to another state (of the finitely many available).

A concrete machine is defined by specifying its behaviour for each combination of the observed symbol and current state, and a *halting state*, upon entering which the machine turns off.

Example 25. Consider a machine that, starting on an empty tape (where every cell contains $\sqcup$), prints three 1s and stops.

We can achieve this using four states: q_0 (initial), q_1 , q_2 , and a halting state q_h . The machine starts in q_0 scanning a blank cell, overwrites it with 1, moves right, and switches to q_1 . In q_1 , it writes another 1, moves right, and switches to q_2 . Finally, in q_2 , it writes a third 1 and enters the halting state q_h :



Here, a label $\sqcup \rightarrow 1, R$ means “if reading $\sqcup$, write 1 and move right”, while S means “stay put”.

Formally, we define a Turing machine as follows.

Definition 26. A *Turing machine* (TM) is a tuple of six components:

$$M = \langle Q, \Sigma, \Gamma, \delta, q_0, q_h \rangle$$

where:

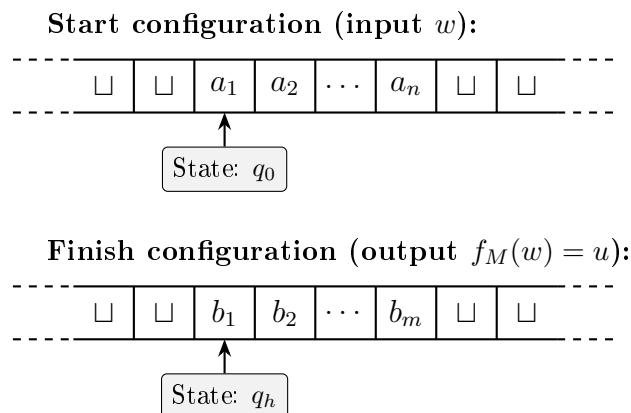
- Q is a finite, non-empty set of **states**;
- Σ is an **input alphabet**;
- $\Gamma \supseteq \Sigma$ is a **tape alphabet** with a special **blank symbol** $\sqcup \in \Gamma \setminus \Sigma$;
- $\delta: Q \times \Gamma \rightarrow Q \times \Gamma \times \{L, R, S\}$ is the **transition function**;
- $q_0 \in Q$ is the **initial state**;
- $q_h \in Q$ is the **halting state**;

The webpage <https://turingmachine.io/> contains several examples and visualisations of Turing machines, and allows you to program your own. Try it!

3.3 Computing functions and deciding languages on Turing machines

A Turing machine M computes a partial function $f_M: \Sigma^* \rightarrow \Sigma^*$, defined as follows:

1. Write the input word $w = a_1 a_2 \dots a_n$ in n consecutive cells, with blanks in all other cells.
2. Place M 's head on the first symbol of w (or on a blank cell if $w = \varepsilon$).
3. Start M from its initial state q_0 and let it run:
 - if M ever reaches the halting state (i.e. *halts*), and the resulting tape contains a word $u \in \Sigma^*$ with all other cells blank, then $f_M(w) = u$, provided that the head is observing the first symbol of u when $u \neq \varepsilon$, or a blank cell when $u = \varepsilon$;
 - in all other cases (no halting, symbols not from $\Sigma \cup \{\sqcup\}$ on the tape, more than one word on the tape), $f_M(w)$ is undefined.



Here $u = b_1b_2 \dots b_m$; when $u = \varepsilon$, the finish configuration contains only blanks.

Clearly, if $\{0, 1\} \subseteq \Sigma$, we can program a machine M computing some function $f_M: \Sigma^* \rightarrow \{0, 1\}$. This machine will classify words of Σ^* into two classes and **decide** the language $\{w \in \Sigma^* \mid f_M(w) = 1\}$, as we have seen with programs before.

Decide-by-state version. Consider an equivalent definition of a Turing machine, specifically designed to decide languages. Instead of one halting state it has two: an **accepting state** q_a and a **rejecting state** q_r . Run such a machine on a tape with a word w —if it ever halts, look at the state: q_a means $f_M(w) = 1$, q_r means $f_M(w) = 0$. If it never halts, we will assume that $f_M(w)$ is undefined.

It is widely accepted that Turing machines capture our intuitive notion of an algorithm.

The Church–Turing Thesis. *Every function that can be computed by an effective procedure (in the intuitive sense: by a human following explicit rules, or by any physical computing device) can be computed by a Turing machine.*

This statement is not a mathematical theorem and cannot be proved formally. It says that Turing machines provide a good formal definition of the informal notion of an algorithm.

What gives us confidence in the thesis is the remarkable convergence of different models of computation. In the 1930s, several other logicians independently proposed their own definitions of computability: Alonzo Church formulated the λ -calculus (we will see it in Lecture 5), Kurt Gödel and Jacques Herbrand developed recursive functions, and Emil Post introduced production systems. Despite their very different styles, all of these models turned out to compute exactly the same class of functions. Later formalisations—such as Markov normal algorithms (we will meet them in Exercise Session 4), register machines, and modern programming languages—likewise proved equivalent.

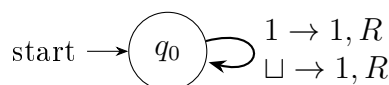
3.4 Examples

As before, we visualise Turing machines as directed graphs where vertices are states and edges represent transitions. A transition $\delta(q, a) = (q', b, D)$ is drawn as an edge from q to q' labelled

$$a \rightarrow b, D$$

meaning: “read a , write b , move in direction D ”.

Example 27. Consider a machine with one work state q_0 over $\Sigma = \{1\}$ and $\Gamma = \{1, \sqcup\}$:

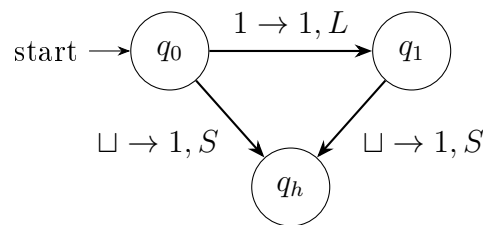


On any input word w , the machine scans rightward indefinitely without ever halting, filling the tape with 1s.⁶

Example 28 (Unary +1). We represent a natural number $n \in \mathbb{N}$ in unary as a sequence of n ones (with 0 represented by the empty word ε).

Given n consecutive 1s on the tape, we wish to produce $n + 1$ consecutive 1s. Rather than walking to the right end of the block, the machine can write 1 to the left of the input: if the head reads 1, it steps one cell left onto the adjacent blank, writes 1, and halts. If the input is empty (the head reads $\sqcup$), it writes 1 in place and halts. In both cases, the machine halts observing the first 1 of the output.

Let $Q = \{q_0, q_1, q_h\}$ with initial state q_0 and halting state q_h :



Example 29 (Unary addition). Consider computing the sum of two numbers $m, n \geq 1$ given as two blocks of ones separated by a + symbol:

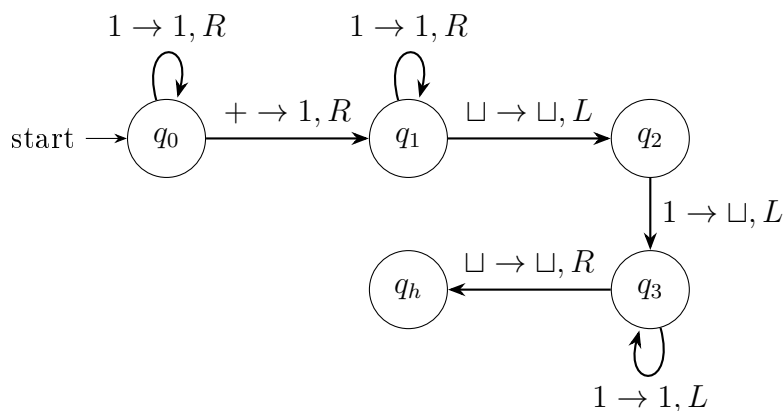
$$\dots \sqcup \underbrace{11\dots 1}_m + \underbrace{11\dots 1}_n \sqcup \dots$$

The procedure is as follows:

1. Walk right across the first block of 1s until reaching the + symbol.
2. Replace + with 1, joining the two blocks into a single block of $m + 1 + n$ ones.
3. Walk to the right end of this block, step back onto the rightmost 1, and erase it (write $\sqcup$), leaving $m + n$ ones.
4. Walk left across all 1s until reaching the blank cell $\sqcup$ to the left of the block, then step one cell right onto the first 1 and halt.

Here is the machine $Q = \{q_0, q_1, q_2, q_3, q_h\}$ with initial state q_0 and halting state q_h :

⁶Strictly speaking, we should also specify a halting state q_h —by definition, every machine must have it. But in this machine, it is *unreachable*—no sequence of transitions starting from the initial state ever leads to it, so we omit it from the diagram.

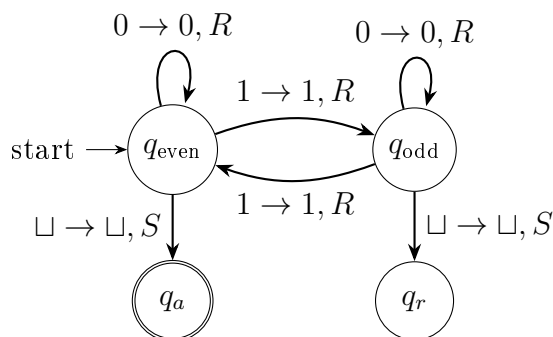


Example 30 (Parity checker). Let us construct a machine with accepting and rejecting states that decides whether a binary string $w \in \{0, 1\}^*$ contains an *even* number of 1s.

We use two work states to track the parity of the number of 1s seen so far: q_{even} (initial state) and q_{odd} . While scanning w , reading 0 keeps the state unchanged, while reading 1 toggles between q_{even} and q_{odd} .

When the machine hits the blank cell $\square$ at the end of the input word:

- from q_{even} , it transitions to the accepting state q_a ;
- from q_{odd} , it transitions to the rejecting state q_r .



3.5 The universal Turing machine

A Turing machine is a single-purpose device: its behaviour on every input is fixed by its transition function. One machine computes binary addition; another decides whether a binary word represents an even number. A computer, however, is more versatile: we program it to perform different tasks. This familiar idea was far from obvious in the 1930s, and we owe its mathematical formulation to Alan Turing.

Turing imagined a universal machine that can be programmed to simulate any other Turing machine. Before we define it, let us remove an apparent complication: different Turing machines may have different tape alphabets.

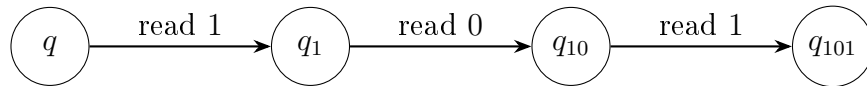
Simulating an alphabet in binary. Let M have tape alphabet Γ . Choose k such that $2^k \geq |\Gamma|$, and assign a different block of k bits to every symbol of Γ . For example, if

$\Gamma = \{\sqcup, a, b, c, d, e, 0, 1\},$	symbol	$\sqcup$	a	b	c	d	e	0	1
	code	000	001	010	011	100	101	110	111

then three cells of M are represented by three blocks on a binary tape:

a	e	c
0 0 1	1 0 1	0 1 1

The simulating machine reads one block at a time. For every state q of M , it has separate states $q_1, q_{10}, q_{101}, \dots$ that record both q and the prefix read so far. While reading the block 101, for instance, it follows:



In state q_{101} , the simulator knows both that M is in state q and that the scanned symbol is e . It can therefore apply the transition prescribed for (q, e) . Further intermediate states remember the new symbol and direction while the simulator rewrites the three bits and moves to the appropriate neighbouring block.

For k fixed in advance, there are only finitely many positions within a block and finitely many prefixes to remember, so all this information fits into finitely many states. We may therefore replace any fixed tape alphabet by a binary one (note that even the blank symbol $\sqcup$ can be encoded as k consecutive 0s, and thus not needed—an empty tape is just a tape filled with 0s).

Consequently, we may freely use finitely many auxiliary tape symbols. For example, we can mark a symbol a by replacing it with a decorated copy $\hat{a}$, or add separators and flags: these are merely extra symbols, and a larger fixed block size encodes them all in binary.

Encoding a machine. It now suffices to encode machines whose tape alphabet is $\{0, 1\}$, with 0 serving as the blank symbol. Suppose such a machine M has m states. Number them

$$q_0, q_1, \dots, q_{m-1},$$

renaming if necessary so that q_0 is initial and q_{m-1} is halting. Begin the code with m ones followed by a zero. From this prefix, a decoder learns m and hence the fixed number of bits needed to write any state number.

Next list the two transitions from each state, first the one for reading 0 and then the one for reading 1. A transition record contains the fixed-width binary number of the new state, the bit to write, and two bits for the direction (say, 00 for L , 01 for R , and 10 for S). There are exactly $2m$ such records, all of the same known length. Their concatenation after the prefix is therefore a binary word which unambiguously encodes the machine M . We will denote this word by $[M]$.

The input w is already binary. We encode the pair consisting of M and w , for example, as a word $\langle [M], w \rangle$, where $[M]$ is the binary word above, while $\langle, \rangle$ and the comma are special symbols that can also be encoded in binary. The particular convention is unimportant; what matters is that a fixed Turing machine can decode it and determine where $[M]$ ends and w begins.

Simulating an encoded machine. A **universal Turing machine** U receives $\langle M, w \rangle$ and maintains three pieces of information on its tape:

- the description $[M]$;
- the current state of the simulated machine;
- an encoding of its tape, with one marked cell (say $\hat{0}$ instead of 0, or $\hat{1}$ instead of 1) indicating the position of the head of M .

Initially these contain $[M]$, q_0 , and the input w . To simulate one step, U reads the marked cell, scans $[M]$ for the transition with the current state and symbol, writes the new symbol, moves the marker to the neighbouring cell, and records the new state. If the marker moves beyond the represented part of the tape, U adds a cell containing the blank symbol 0. Each operation is a finite combination of scans, comparisons, and copies, so a fixed Turing machine can perform it. Repeating this procedure simulates every step of M .

Theorem 31 (Turing, 1936). *There exists a Turing machine U such that, for every Turing machine M and input w ,*

$$U(\langle M, w \rangle) = M(w).$$

That is, U halts exactly when M halts, and in that case produces the same output.

The universal machine is the theoretical model of a general-purpose computer: $[M]$ is a program, w is its input, and U executes the program. We can now restate Theorem 7 for Turing machines:

Theorem 32 (Turing, 1936). *The language*

$$\text{HALT}_{\text{TM}} = \{\langle M, w \rangle \mid M \text{ halts on } w\}$$

is undecidable.

Lecture 5. Lambda calculus

So far, we have emphasised that programs can be treated as data. Now, we will study another formal definition of an algorithm, historically one of the earliest, where the tables are turned. In Alonzo Church's lambda calculus, everything is a program.

5.1 Functions, in math and in lambda calculus

We are used to defining functions by writing $f(x) = x + 1$, or, sometimes, by $f: x \mapsto x + 1$. Applying f to 4 gives 5: $f(4) = 5$.

In lambda calculus, the same function is denoted by $\lambda x.x + 1$ —literally, “a function that takes x and returns $x + 1$ ”.⁷ Application goes like this:

$$(\lambda x.x + 1)4 = 4 + 1 = 5$$

This is not pure lambda calculus: to keep this first example simple, we use numbers and the notation $+1$ as if they were defined. Soon we will construct these things, and many more.

A function may return another function. For example,

$$(\lambda x.\lambda y.x + y)3 = \lambda y.3 + y$$

Applying the result to 4 gives 7. Thus, a function of many arguments can be represented by a function receiving the first argument and returning a function of the rest of the arguments, with the first one fixed. This is called **currying**.⁸

Definition 33. A *lambda term* is a variable x , an application (MN) of one term to another, or an abstraction $(\lambda x.M)$.

Application associates to the left: MNP means $(MN)P$; the parentheses around terms are omitted where unnecessary. The body of an abstraction extends as far to the right as possible: $\lambda x.MN$ means $\lambda x.(MN)$.

5.2 Computation by substitution

The basic computation rule is **beta reduction**.⁹

$$(\lambda x.M)N \longrightarrow_{\beta} M[x := N]$$

Read this as: replace the occurrences of the parameter x in the body M by the argument N . It may be performed inside a larger term, including inside the body of an abstraction. We write $\longrightarrow_{\beta}^*$ for zero or more applications of beta reduction.

⁷Why λ ? Church never explained clearly. Wikipedia's page on lambda calculus contains some details.

⁸Chicken curry is not relevant here, Haskell Curry is.

⁹Why β ?! Because there is a simpler one, called α , so it's just numbering (and Church's love for Greek letters).

Example 34. Let $ID = \lambda x.x$ and $K = \lambda x.\lambda y.x$. Then:

$$IDA \rightarrow_{\beta} A \quad KAB \rightarrow_{\beta} (\lambda y.A)B \rightarrow_{\beta} A$$

The first function returns its argument; the second keeps its first argument and discards the second.

A variable occurrence is **bound** if it is introduced by a surrounding λ ; otherwise it is **free**. In $\lambda x.xy$, the occurrence of x is bound and that of y is free. A term with no free variables is **closed**. Substitution replaces only free occurrences: in $\lambda x.x$, the x is bound, it is a local name for $\lambda x.x$. So:

$$(\lambda x.x(\lambda x.x))A \rightarrow_{\beta} A(\lambda x.x)$$

The name of a local parameter does not matter: $\lambda x.x$ and $\lambda z.z$ describe the same function. Consistently renaming a bound variable to a fresh name, meaning one not already occurring in the term, is called **alpha conversion**: $\lambda x.x \equiv_{\alpha} \lambda z.z$. We will identify terms that differ only by such renaming.

Mind the bounds! The term $K = (\lambda x.\lambda y.x)$ of Example 34 says: "take x , remember it, then take whatever and return x ". Imagine we apply it to y : $(\lambda x.\lambda y.x)y$. A 'dumb' reduction substitutes y for x in the body and produces $\lambda y.y$: we were supposed to remember the first argument (which was y), but it is now lost! Indeed, $\lambda y.y$ will take an argument and output *that argument*, not the term y passed as the first argument to K . So this is a wrong way of doing beta reductions. The correct way is: when substituting M for the free occurrences of x in N , first rename any bound parameters of N that would otherwise bind free occurrences of variables in the inserted M , using names fresh for both terms:

$$(\lambda x.\lambda y.x)y \equiv_{\alpha} (\lambda x.\lambda z.x)y \rightarrow_{\beta} \lambda z.y$$

Now the answer is a constant function returning the external y , and we are happy. This convention is called **capture-avoiding substitution**.

5.3 Functions as arguments of other functions

Function composition is itself a function:

$$COMP = \lambda f.\lambda g.\lambda x.f(gx)$$

What does it do? It takes the first argument, f , then the second, g , and returns a function that takes x and applies g to x , then f to $g(x)$ (in standard math notation), then returns the result $f(g(x))$. Indeed, $COMPFGX \rightarrow_{\beta}^* F(GX)$.

Example 35. Let $DBL = \lambda f.\lambda x.f(fx)$, the operation that applies a function twice. Then

$$DBLFX \rightarrow_{\beta}^* F(FX) \quad DBL(DBLF)X \rightarrow_{\beta}^* F(F(F(FX)))$$

Note the parentheses. If we write without them, left-associativity will enforce another reading: $DBLDBLFX$ is $((DBLDBL)F)X$. In this case, however, the result is the same. Apply the definition of the *outer* DBL first: it gives $DBL(DBLF)X$, so again F is applied four times.

5.4 Data as functions

5.4.1 Booleans

What must a Boolean do? It must choose between two alternatives. Define

$$\text{TRUE} = \lambda x.\lambda y.x \quad \text{FALSE} = \lambda x.\lambda y.y$$

So TRUE is our old friend K, and FALSE is the dual: “Take the first argument, forget it, take the second and return it”.

Then $\text{TRUE } AB \rightarrow_{\beta}^* A$ and $\text{FALSE } AB \rightarrow_{\beta}^* B$. A conditional needs no additional mechanism:

$$\text{IF} = \lambda p.\lambda x.\lambda y.pxy$$

That is, $\text{IF } ABC$ is the λ -way of writing “if A is true, then B , else C ”. Indeed:

$$\text{IF TRUE } BC \rightarrow_{\beta}^* \text{TRUE } BC \rightarrow_{\beta}^* B$$

and

$$\text{IF FALSE } BC \rightarrow_{\beta}^* \text{FALSE } BC \rightarrow_{\beta}^* C$$

To negate a Boolean, exchange its two alternatives:

$$\text{NOT} = \lambda p.\lambda x.\lambda y.pyx$$

For example,

$$\text{NOT TRUE} \rightarrow_{\beta} \lambda x.\lambda y.\text{TRUE } yx \rightarrow_{\beta}^* \lambda x.\lambda y.y = \text{FALSE}$$

Conjunction can be implemented by

$$\text{AND} = \lambda p.\lambda q.pqp$$

That is, if p is true, return q ; otherwise return p , which is already false.

5.4.2 Pairs

To store two values, make a function that passes both to whoever wants to use them:

$$\text{PAIR} = \lambda a.\lambda b.\lambda f.fab \quad \text{FST} = \lambda p.p \text{ TRUE} \quad \text{SND} = \lambda p.p \text{ FALSE}$$

Thus $\text{FST } (\text{PAIR } AB) \rightarrow_{\beta}^* A$, and similarly for the second component.

Embrace the λ -philosophy: a pair is a function; to get its first or second value, *apply it* to the right argument. If you want a sum of the two components, apply your pair to the function of addition: $(\lambda f.fab) \text{ ADD} \rightarrow_{\beta} \text{ADD } ab$.

Wait-wait, how do we define ADD? And what is a sum, if we haven’t seen any numbers so far, only terms, terms, terms... Here we go.

5.4.3 Natural numbers

If the only thing we can do is apply functions, how can we simulate a number $n \in \mathbb{N}$? Well, the number n can tell us to apply a function n times. Define the **Church numerals**

$$\bar{0} = \lambda f.\lambda x.x \quad \bar{1} = \lambda f.\lambda x.fx \quad \bar{2} = \lambda f.\lambda x.f(fx) \quad \dots$$

In general, $\bar{n} = \lambda f.\lambda x.f^n(x)$, where $f^0(x) = x$ and $f^{n+1}(x) = f(f^n(x))$. The superscript here is an abbreviation for repeated application. In particular, DBL from Example 35 is $\bar{2}$.

To obtain $n + 1$, the successor of n , perform its repetitions and apply f once more. That is, $\overline{n+1}$ can be represented as $\lambda f.\lambda x.f(\bar{n}fx)$. Literally: “take f , take x , apply f to x exactly n times, and then once more”.

So the function that takes $\bar{n}$ and returns $\overline{n+1}$ is then:

$$\text{SUCC} = \lambda n.\lambda f.\lambda x.f(nfx)$$

For example,

$$\text{SUCC} \bar{2} \longrightarrow_{\beta}^* \lambda f.\lambda x.f(f(fx)) = \bar{3}$$

To add m and n , apply successor m times, starting at $\bar{n}$:

$$\text{ADD} = \lambda m.\lambda n.m \text{ SUCC } n$$

A Church numeral can iterate any function, including a function on Church numerals. Here $\bar{m}$ receives SUCC as the function to repeat and $\bar{n}$ as the starting value. For example,

$$\text{ADD} \bar{2} \bar{3} \longrightarrow_{\beta}^* \text{SUCC}(\text{SUCC} \bar{3}) \longrightarrow_{\beta}^* \bar{5}.$$

For multiplication, add n a total of m times, starting at zero:

$$\text{MULT} = \lambda m.\lambda n.m (\text{ADD } n) \bar{0}$$

Note the parentheses! $\text{ADD } n$ is a function that takes a numeral and adds n to it. Repeating this function m times from $\bar{0}$ gives $\bar{n}, \bar{2n}, \dots, \bar{mn}$; if $m = 0$, the result is simply $\bar{0}$.

Remark 36. Note that $\bar{0}$ and FALSE are the same term up to parameter names.

5.5 Halting and nontermination

What would be the λ -counterpart of our `never_stop` program? Of the Turing machine that runs forever to the right?

A term is in **normal form** if it contains no subexpression of the form $(\lambda x.M)N$. For example, $\lambda x.x$ and all Church numerals are normal forms. A term *has* a normal form if some finite sequence of reductions reaches one.

Let

$$\Delta = \lambda x.xx \quad \Omega = \Delta\Delta$$

Then $\Omega \rightarrow_{\beta} \Omega \rightarrow_{\beta} \dots$. Its only possible step reproduces the same term, so it has no normal form. Notice that Δ itself is a normal form.

Different subexpressions may be ready to reduce:

$$(\lambda x.ID) \Omega \rightarrow_{\beta} ID \quad \text{and} \quad (\lambda x.ID) \Omega \rightarrow_{\beta} (\lambda x.ID) \Omega$$

That is, reducing the outer application gives ID immediately. Reducing inside the argument can continue forever. Thus, the existence of an infinite reduction does *not* prove that a term has no normal form.

Theorem 37 (Church–Rosser). *If $M \rightarrow_{\beta}^* P$ and $M \rightarrow_{\beta}^* Q$, then there is a term R such that $P \rightarrow_{\beta}^* R$ and $Q \rightarrow_{\beta}^* R$, allowing renaming of bound variables (alpha conversions).*

This property is called **confluence**. In particular, a term cannot have two different normal forms: neither could reduce further to a common result. Confluence does not say that every choice of reductions terminates.

5.6 Trying it in Scheme

Scheme is a *functional programming language* in which functions can be passed around just as in our examples. An online interpreter, such as <https://try.scheme.org/>, lets us try:

```
(define dbl (lambda (f) (lambda (x) (f (f x)))))
(define succ (lambda (x) (+ x 1)))

(( dbl succ) 10)           ; 12
((( dbl dbl) succ) 10)    ; 14
```

Here `define` names a value, `(f x)` applies it, and semicolons introduce comments. Scheme supplies built-in numbers and evaluates arguments before function calls.